

Full Length Article

Mapping of residual stresses in as-built Inconel 718 fabricated by laser powder bed fusion: A neutron diffraction study of build orientation influence on residual stresses



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ABSTRACT

Manufacturing of functional (ready to use) parts with the powder bed fusion method has seen an increase in recent times in the field of aerospace and in the medical sector. Residual stresses (RS) induced due to the process itself can lead to defects like cracks and delamination in the part leading to the inferior quality of the part. These RS are one of the main reasons preventing the process from being adopted widely. The powder bed methods have several processing parameters that can be optimized for improving the quality of the component, among which, build orientation is one. In this current study, influence of the build orientation on the residual stress distribution for the Ni-based super-alloy Inconel 718 fabricated by laser-based powder bed fusion method is studied by non-destructive technique of neutron diffraction at selected cross-sections. Further, RS generated in the entire part was predicted using a simplified layer by layer approach using a finite element (FE) based thermo-mechanical numerical model. From the experiment, the part printed in horizontal orientation has shown the least amount of stress in all three directions and a general tendency of compressive RS at the center of the part and tensile RS near the surface was observed in all the samples. The build with vertical orientation has shown the highest amount of RS in both compression and tension. Simplified simulations results are in good agreement with the experimental value of the stresses.

1. Introduction

Additive manufacturing (AM) is on the rise for manufacturing parts with complex geometrical features. It has come a long way from being used for prototyping to serial production [1]. However, AM of metallic parts for serial production has not yet been fully established due to the complex nature of the process and the materials. Out of many different methods for manufacturing parts on a layer-by-layer basis, the laser-based powder bed fusion method (L-PBF) is one of the most widely used techniques mainly due to the range of material and process parameter selection available. Traditionally, the manufacturing process has limited the design freedom, i.e. the manufacturing process is one of the key envelopes for the design, and anything out of this design envelope would not be possible to manufacture using that process of interest. This approach often results in less efficient parts and increases the

number of parts in an assembly as well. AM brings a new dimension to the design philosophy through more design freedoms. Consequently, parts with complex geometry can be designed and manufactured with a goal to reduce weight and to consolidate the parts (minimizing assembly work) [2–4] without compromising the mechanical integrity. AM not only allows for producing complex parts but is also eco-friendlier and more sustainable if properly implemented. For example, in powder bed methods the remaining powder can be recycled and reused in combination with fresh powder without compromising much of the quality [5,6]. Also, AM can be used to repair and retrofit old parts manufactured with traditional methods [7–9]. A variety of materials have been developed and investigated for the purpose of AM. One material grade that is of high importance for the energy and aerospace industries is nickel-based super-alloys due to their good mechanical properties at elevated temperatures. These nickel-based super-alloys are

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often used in aero and gas turbine engines especially for high-pressure turbine blades and hot section components. For such components, traditional manufacturing technologies are cost-ineffective and highly time-consuming [10,11] when compared to AM.

In AM, processing parameters are intertwined with each other and have a significant influence on the outcome of the parts in terms of properties such as density, porosity, mechanical strength, etc. It is necessary to fine-tune these parameters for each alloy in order to achieve the desired mechanical characteristics. Several studies have been performed to relate processing parameters like laser power and scan speed/strategy [12–17] to the final part outcome in terms of density, roughness, residual stresses, etc. Among several difficulties which AM of metals has to overcome before its wide implementation, control of residual stresses (RS) is one important factor. These RS are self-balanced stresses that originate due to the incompatibility of the layers, mainly attributed to the temperature gradient between different layers during printing in powder bed processes [18]. RS can lead to both superior or inferior mechanical properties [19–21]. They are of particular interest in AM processes because a high level of residual stresses can lead to crack formation in the part while printing causing heavy warping from the base plate [18,22]. From the mechanical point of view, compressive residual stresses near the surface are known to increase fatigue life by increasing the resistance against crack formation and propagation. On the other hand, tensile residual stresses promote the development of fatigue damage and may lead to the failure of parts at very low external loads. There are ways to reduce/modify the RS in the manufactured parts by using post-processing techniques like heat treatment, shot peening, machining, etc. [19,23]. However, for parts with complex geometrical features, these surface treatment methods might not be feasible. Most of the previous works with regard to RS are mainly focused on scan strategy, laser power, and laser speed, etc., while the influence of build direction on RS has only been investigated in very few works previously [24–27]. Furthermore, most of the previous works on the influence of the process parameters dealt with the outcome in terms of mechanical properties and were conducted with a simple geometry without any significant change in cross-section dimensions during printing [28–31]. A need to understand the influence of the build orientation arises as parts are getting more and more complex, which in turn makes the RS generated even more complex to manage. Together with other tailored processing parameters for a certain kind of alloy system, proper selection of the build orientation based on the part geometry to minimize the RS generation will be beneficial.

In order to measure RS various methods have been developed which can be placed in two main groups: destructive and non-destructive testing methods. Hole drilling [32] and the contour method [33], etc. are semi-destructive and destructive techniques, while X-ray diffraction, neutron diffraction, and ultra-sonic testing are nondestructive testing [34]. The X-ray diffraction method gives the information regarding the stress state near the surface only due to lack of penetration depth, whereas the neutron diffraction method is capable of measuring RS inside the bulk of a part and thus can be used to characterize residual stress distributions inside the materials. For comparison and benefits of different techniques for RS measurement, readers are referred to the relevant literature in the field of RS measurement technique [34–36]. This work is focused on the use of the non-destructive technique of neutron diffraction for the measurement of RS. In relation to RS measurement with neutron diffraction, various aspects of AM like the influence of the geometry change in simple cylinders, support structure, hatch deposition length, and removal from base plate have been studied previously [37–39]. Also with neutron diffraction, RS in thin-walled geometry in as-built condition together with build plate has been studied [40]. RS measurements on AM parts can help to understand the generation of RS and their correlation to the processing parameters and geometry. In the available publications, RS distributions were obtained for some lines [41] or cross-sections in parts of simple geometry [39]. To the author's best knowledge, more complex

shapes have not been investigated for the RS development with neutron diffraction.

In order to predict the RS generation during the printing, various modeling strategies also have been developed to understand the influence of the major processing parameters like laser speed, laser power, interlayer dwell time, scan strategy, etc. with varying degrees of complexity [16,42–46]. However, most of the works done previously are focused on real process simulation, and for this, extensive knowledge of the physical phenomenon is required, which can be both time and resource consuming. Therefore, a simplified simulation technique for the residual stress prediction in a qualitative manner will be a beneficial tool for a quick check of the influence of part geometry and the influence of other processing parameters.

In the current work, the non-destructive neutron diffraction method and a simplified finite element modeling approach have been employed to investigate residual stresses in L-shaped parts built in three different orientations. The focus is to provide means to identify the suitable building direction for printing the part with the aim to reduce the RS field and to promote a more homogeneous distribution of it.

2. Sample preparation

Samples for this study were built from Inconel 718 using an SLM 125 H L (SLM Solutions Group AG, Germany) machine equipped with a soft rubber re-coater system. The material used for the printing was gas atomized Inconel 718 powder with a particle size range of 10–45 μm provided by the machine manufacturer. In the printing process, a mixture of recycled powder from an older batch and virgin powder (8 kg of the recycled powder and 10 kg of virgin powder) was used. It is a common practice to mix recycled and virgin powders in the industry to minimize the powder waste. The parts were produced with process parameters optimized for Inconel 718 with respect to low porosity. Some of the processing conditions and process parameters can be seen in Tables 1 and 2 respectively. A scanning strategy with stripe pattern and layer rotation of 16 degrees for each layer, together with two borders and one fill contour were used for all samples. The scan strategy can be seen in Fig. 1. The laser movement during printing is complex in nature and it can be divided into 3 different movements namely global, semi-global, and local (see Fig. 2). Global can be defined as the laser movement with respect to the layer and it can be from the top left of the part to the bottom right or the opposite way (see Fig. 2a and b). The semi-global movement is seen as the movement of the laser within a 10 mm strips where it moves from left to right and the local movement is the movement within the strips where the laser moves up and down (see Fig. 2). All these movements of the laser changes with each layer rotation.

L-shaped samples of dimensions of 55 mm $\times$ 10 mm $\times$ 20 mm with a hole of 5 mm in diameter in one of the sides as seen in Fig. 3 were manufactured for the study. The hole present in the sample will act as a possible stress concentrator and provide information about the effect of such a feature. The L-shape with a hole was chosen because when it is manufactured in different orientations it can generate different stress fields and also at the same time will help to study the influence of the features such as corners, short and long sides, holes, etc. Three different build directions, namely horizontally built (HB), vertically built (VB), and built at 45° angle (45B), were used. Six parts in total, i.e. 2 for each

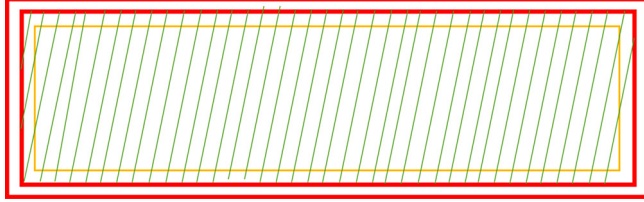
Table 1
General setup for printing.

Parameters	Values
Layer thickness [μm]	30
Build plate temperature ($^{\circ}\text{C}$)	200
Atmosphere	Argon
Build plate	SS316L

Table 2

Process parameter for the printing of different section.

Parameters	Hatching	Border	Fill contour
Laser power (W)	200	100	125
Laser speed (mm/s)	900	450	450
Hatching distance (mm)	0.12	0.08	0.08

**Fig. 1.** Section of scan strategy for printed parts. Borders are red, fill contour orange and hatching green (not to scale).

orientation, were manufactured within the same batch for consistency reasons (Fig. 3) with the same printing parameters as detailed in Tables 1 and 2. Due to the soft rubber lip used in the re-coater, parts were rotated at a 10° angle with respect to the re-coater blade as a common practice. This is done because long edges of the printed parts can damage the rubber lip during the printing process if they are perpendicular to the re-coater lip. The parts were then removed from the build plate using a wire electric discharge machine (W-EDM) (SodickA750, Sodick Group). For each build direction, one sample was used for RS mapping by neutron diffraction and the other was used to prepare stress-free reference samples.

3. Experiments

Three different measurement techniques were used in this study to address different aspects of the RS distributions generated in the AM parts. The neutron diffraction was used to study the RS inside of the bulk of the AM material. The X-ray diffraction measurements were appropriate for the surface stress measurements that are unsuitable for the neutron technique. Additionally, the overall distortions associated with the separation of the samples from the base plate and corresponding stress redistribution were measured by the laser scanning technique.

3.1. Strain measurement by Neutron diffraction

The main idea of measuring strains with the use of neutrons relies on the Bragg law of diffraction (Eq. (1)), where the inter-planar spacing d_{hkl} of a particular set of lattice planes (hkl) is used as a strain gauge. By

diffraction, the inter-planar spacing at the location of interest is measured and then with the help of the stress-free inter-planar spacing, the elastic strain is calculated (Eq. (2)). To derive the corresponding stresses using Hooke's law (Eq. (3)), elastic strains in three orthogonal sample directions are often measured.

$$\lambda = 2d_{hkl}\sin\theta_{hkl} \quad (1)$$

where,

λ - wavelength of the radiation used

d_{hkl} – inter-planer distance measured using hkl planes

θ - incident angle for the radiation

$$\varepsilon = \frac{(d - d_0)}{d_0} \quad (2)$$

where,

ε – elastic strain, d - measured interplanar distance,

d_0 – interplanar distance for stress- free sample

$$\sigma_{ij} = \frac{E_{hkl}}{1 + \nu_{hkl}} \left(\varepsilon_{ij} + \frac{\nu_{hkl}}{1 - 2\nu_{hkl}} \varepsilon_{qq} \delta_{ij} \right) \quad (3)$$

where,

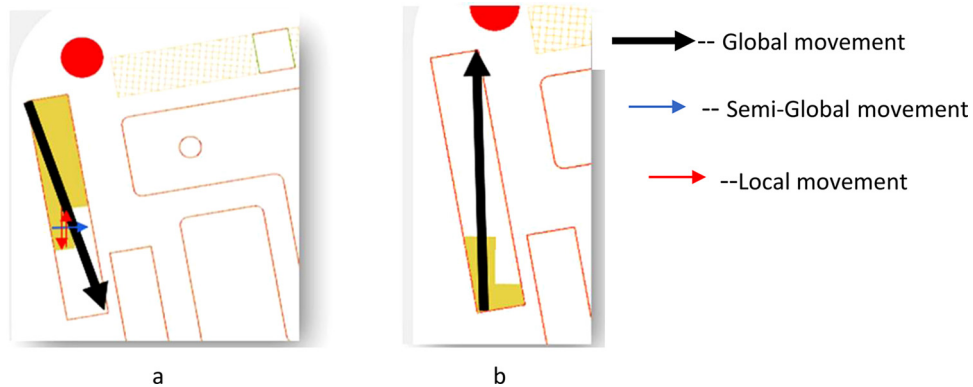
σ_{ij} – stress calculated for a particular direction, E_{hkl} - Elastic coefficient for hkl plane

ν_{hkl} - Poisson's ratio for hkl plane

3.2. Experimental setup at KOWARI

Experimental evaluation of the RS was done at the Beamline KOWARI of the Australian Nuclear Science and Technology Organization (ANSTO). KOWARI [47] is a dedicated strain scanner that was used for the bulk residual stress measurements. A slit size of $2 \times 2 \text{ mm}^2$ was used for the incoming and outgoing beams to provide a $2 \times 2 \times 2 \text{ mm}^3$ gauge volume. A nominal wavelength of $1.5 \text{ \AA}$ was selected as the instrument is optimized for this wavelength in terms of neutron flux and instrumental resolution. With this wavelength, stress measurements for FCC materials like Inconel 718 can be done using the (311) reflections with a 2θ of about 90° to have a cube-shaped neutron gauge volume. The position of each sample on the diffractometer was determined using a surface scan: moving the sample into the neutron gauge to determine the sample surface position from the obtained intensity changes. As a result, the positioning accuracy was usually better than 0.1 mm . The general setup of the sample in the beamline can be seen in Fig. 4.

The obtained neutron diffraction profiles were fitted using the Gaussian model to determine the peak position and therefore the peak shift with an accuracy of $50 \mu\text{strain}$. Elastic strains in the three principal sample directions were obtained according to Eq.(1) and Eq. (2), and the stresses were calculated using Eq. (3) with the elastic constant

**Fig. 2.** Schematic of laser movement during the printing of VB sample a: global laser movement from top left to bottom right and b) laser global movement from bottom to top.

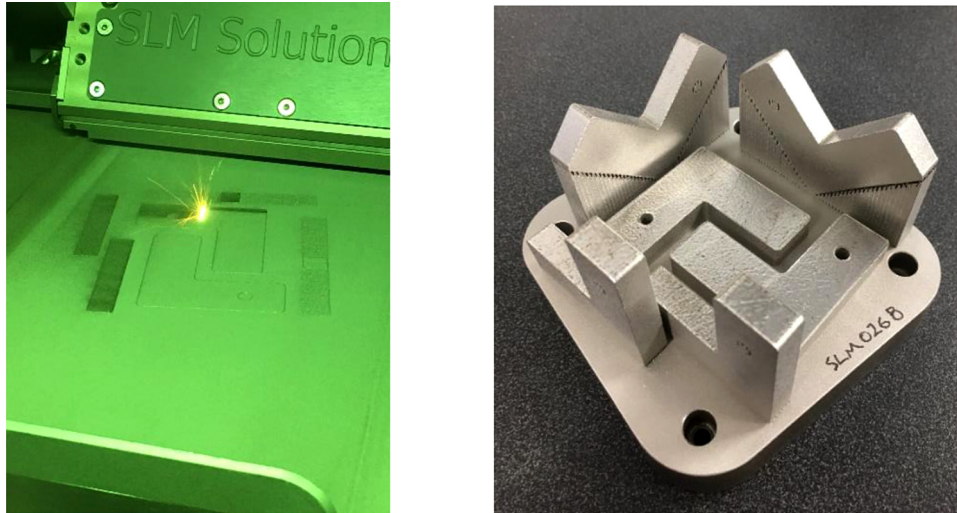


Fig. 3. Samples during and after printing. The hole in the VB sample is on the top part.

values listed in Table 3. In the best possible experiment, shear stress components would also be required to fully characterize the stress state, however, this would require at least twice the measurement time. Thus, due to neutron beamtime limitations, only three normal stress components were measured, but it should be taken into account that they are not necessarily the principal stress components especially in the corner areas (or around other topological features). This approach is commonly used as the normal stress component data normally are sufficient for structural integrity analysis and validation of any FEM calculations.

For each sample, four cross-sections were chosen for 2D strain and stress mapping and were labeled C1, C2, C3, and C4 as shown in Fig. 5. The sections were located with respect to the outer reference sample edges, as shown in Fig. 5b. The measurements in the sections were systematically done for each sample using the same mesh as described below.

In order to avoid apparent (or spurious) strains due to a partially

Table 3

Diffraction constants.

E_{311} [GPa]	202 [51]
ν_{311}	0.31 [51]
E_{220} [GPa]	280 [19]
ν_{220}	0.298 [19]

illuminated neutron gauge volume [48,49], all the measurement positions were at least 1.6 mm below all surfaces. For the cross-sections C1, C2 and C4, each line was scanned with a step size of 2.1 mm in the Y-direction/ Z-direction (for C4) and 1.7 mm in the X-direction. For the cross-section C3, the step size was increased to 2.85 mm in the Z-direction and 1.7 mm in the X-direction. This was done to save time during the experiment as well as to avoid the hole while measuring the sample. Measurement meshes for the respective cross-sections are

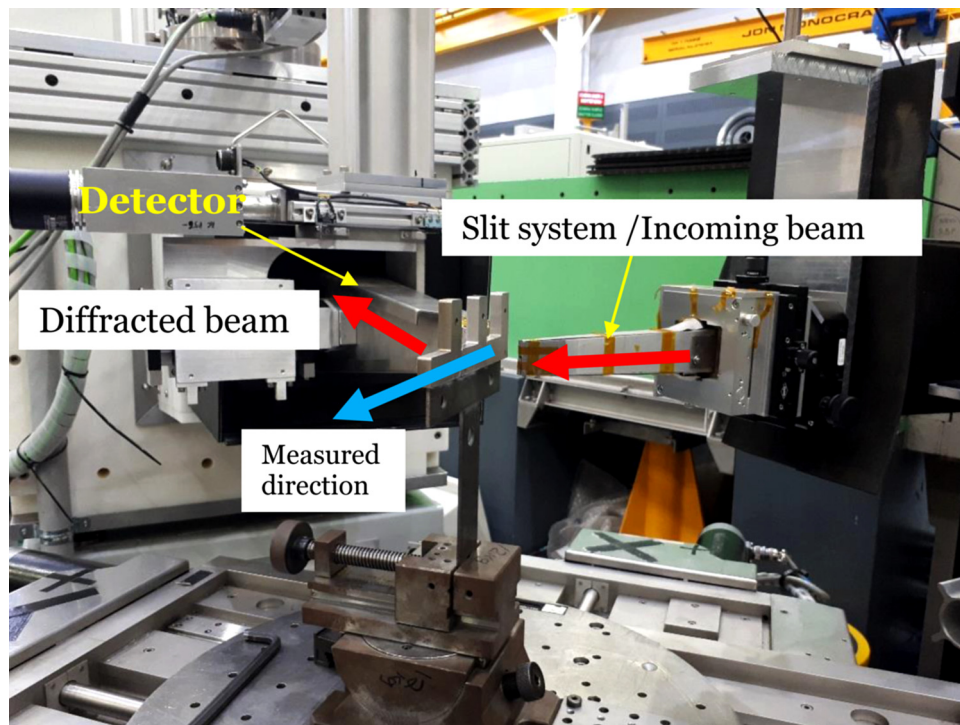


Fig. 4. Measurement setup at KOWARI.

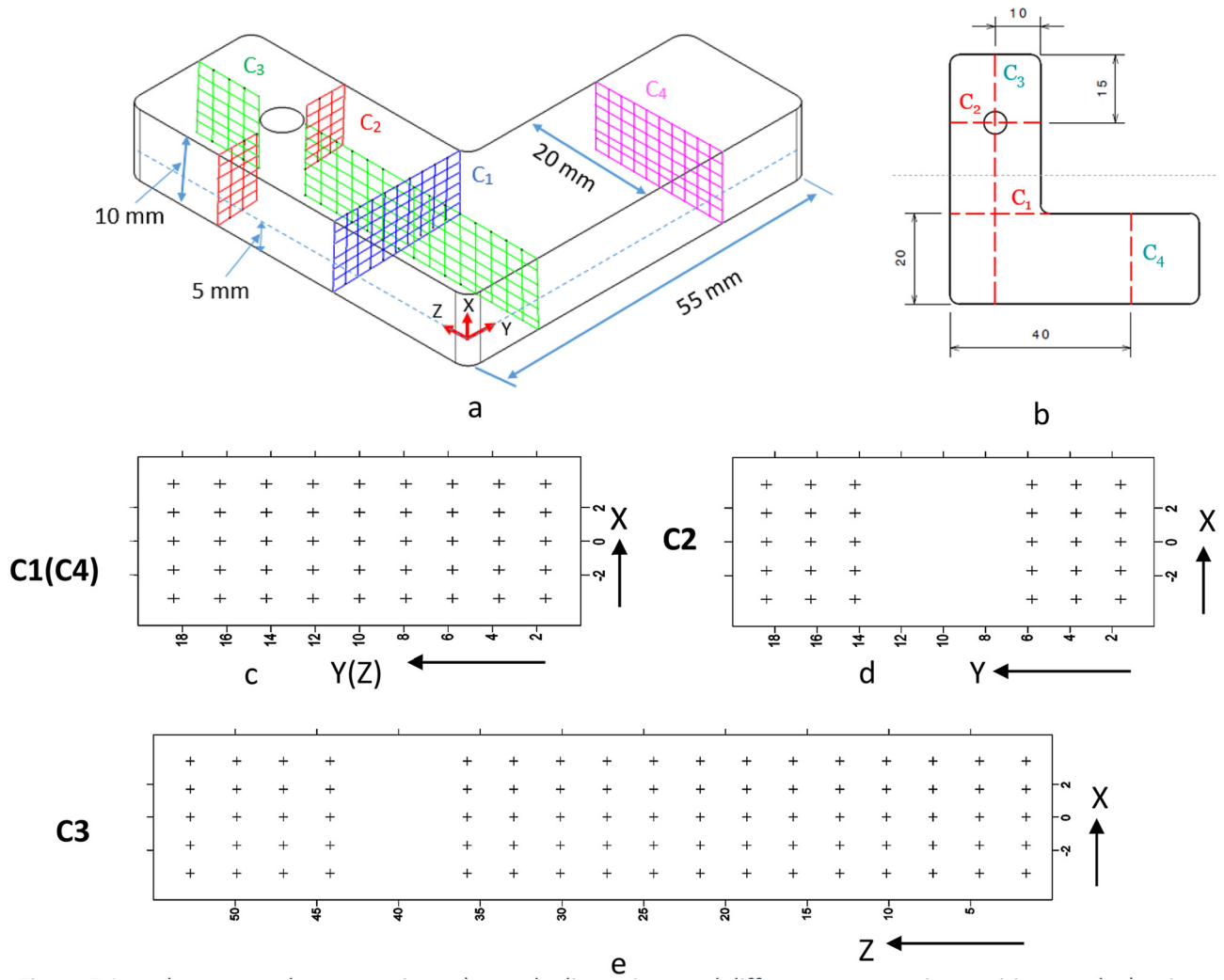


Fig. 5. Sample measured cross-sections a) sample dimensions and different cross-section positions, c, d, e) points measured at different cross-section.

shown in Fig. 5c-e.

Measurements of strains in both the Z and Y-directions were performed in transmission mode while in reflection mode for the X-direction. To avoid strong beam attenuation, after the strains in the first half-thickness were measured, the sample was rotated by 180° to complete the measurement in the other half-thickness.

In total 205 points were measured for each sample in each of three principal directions and the average time for the measurement for each point was 5 min.

3.3. Reference sample (d_0 sample)

Due to time limitations for the measurements, the stress-free interplanar spacing was experimentally evaluated for the vertically built sample only. By electric discharge machining (EDM) a 3 mm thick sample was extracted and 3 mm × 3 mm × 3 mm cubes were further cut according to Fig. 6. Neutron diffraction measurements were performed along the three principal sample directions.

3.4. Surface stress measurements

From the neutron diffraction experiments, stress values can be obtained no nearer the surface than a 1.6 mm and in order to have a complete set of the stress distribution, stresses close to the surface were measured using a lab-scale X-ray diffractometer (Seifert System, Germany) with Cr-K α radiation source (wavelength of 2.29 Å). As the γ -

(311) is outside the measurement range by Cr-K α , the γ -(220) reflection at 2 θ ~ 128.5° was measured instead. Upon the assumption of a plane stress condition, the $\sin^2\psi$ method was used with 11 ψ angles (see Fig. 7) between +55° to -55°. The stress, σ_φ , is then calculated based on the following equation [35]:

$$\frac{d_\psi - d_0}{d_0} = \frac{1 + \nu_{hkl}}{E_{hkl}} \sigma_\varphi \sin^2\psi - \frac{\nu_{hkl}}{E_{hkl}} (\sigma_{11} + \sigma_{22}) \quad (4)$$

where,

σ_φ - stress measured in the in-plane direction φ , E_{hkl} - Young's modulus for hkl plane, ν_{hkl} - Poisson's ratio for hkl plane

ψ - Angle of tilting from the sample surface normal towards the direction φ ,

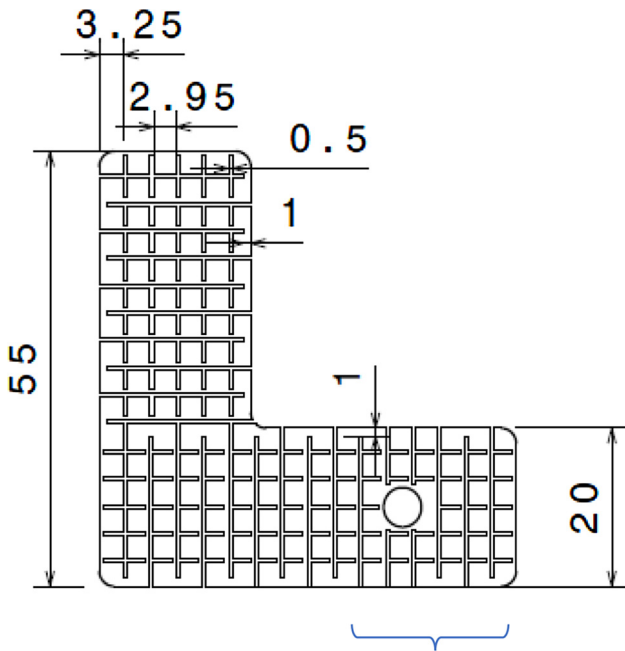
d_ψ - d-spacing measured at tilt angle ψ [Å]

d_0 - stress-free d-spacing [Å]

Stress (σ_φ) was calculated from the slope of the linear fit of ' d_ψ vs $\sin^2\psi$ ' plot and d_ψ measured at $\psi = 0^\circ$ is used as approximated d_0 . Uncertainties in stress value measured were evaluated based on standard deviation from the linear fit. The values of the diffraction elastic constants used for the calculation are listed in Table 3. The stress was measured at the locations shown in Fig. 8.

3.5. Displacement measurements by 3D surface scanning

After removal from the baseplate, the samples were 3D-scanned using a GOM Atos Triple Scan III to measure the distortions after cut off



Measured area for d_0 values

Fig. 6. d_0 reference sample.

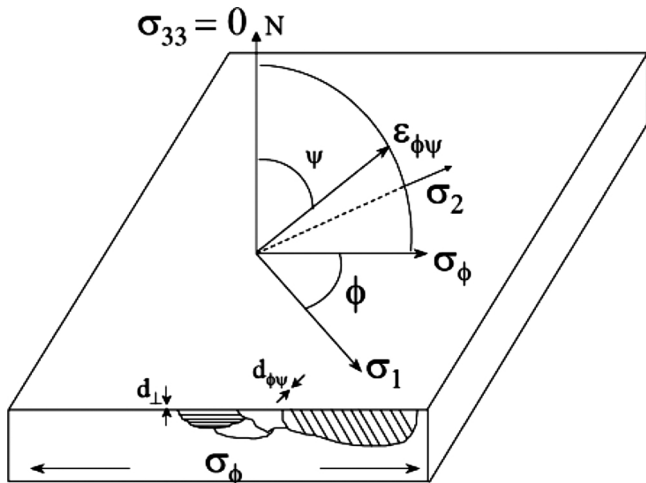


Fig. 7. General schematic RS determination using $\text{Sin}^2\psi$ method in lab based XRD. Both σ_1 and σ_2 lie in the plane of the specimen surface [50].

Points for the Surface measurement

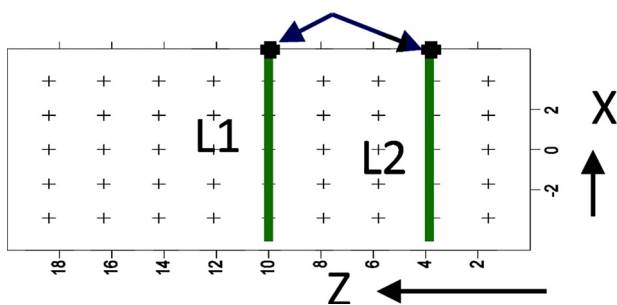


Fig. 8. Points selected for surface stress measurements for C4 section.

from the base plate and a "Best fit" has been used to measure the overall displacement of the parts with respect to the CAD file. A "Best-fit" can be defined as the best of all averages in deviation from the nominal and utilizes the sum of the least squares best fit algorithm. Thus, it iterates all the surfaces referred to in the comparison to create the minimum overall deviation in relation to the nominal. These measurements were used to relate the difference in residual stress measured for different orientations.

4. Simulation

Simulating the real process of powder bed fusion AM process with a moving heat source is very complex and requires a long computational time. As the aim was to predict critical areas with respect to RS in a very short time for complex geometries, a simplified fully coupled thermo-mechanical finite element model for the simulation of the build process and the RS generation was developed using the commercial software ABAQUS [52]. As the temperature gradient between the subsequent layers is one of the main factors contributing to the RS generation in the process, in this simplified approach it was assumed that the RS starts to form when the material starts to solidify and a temperature close to the melting temperature was used as the load applied to the part. The choice of the starting temperature depends upon the material being investigated.

The general workflow of the model is shown in Fig. 9. The part along with the base plate is initially sectioned in multiple layers (combined layers). In the current simulation, each combined layer consists of 10 real powder layers being applied for the print. Initially the combined layers are deactivated and are re-activated layer-by-layer with a certain amount of time for temperature loading followed by cooling. Time for re-activation and subsequent cooling is calculated from the laser scanning pattern for combined layers and the movement of powder bed and re-coater, respectively. For heat dissipation, boundary conditions for convection are applied for the part sections that are submerged into the powder bed instead of modelling heat conduction through the powder bed. Also, radiation was prescribed to the top surface of each layer. Finally, when the part was fully activated with all layers and cooled down to room temperature, the base plate was deactivated to simulate separation of the part from the base plate and mechanical boundary conditions were imposed on the part to prevent any rigid body motion during this deactivation of the base plate. As for the material model, an elastic-plastic model with isotropic hardening behavior was used with temperature-dependent material data according to Tables 4 and 5, respectively. A detailed description of the finite element model will be presented in a separate paper.

5. Results

5.1. Displacement after removal from the base plate

Plots in Figs. 10–12 show the overall displacement of the surface of

Table 4

Material properties of In718 [53].

Temp. [K]	Thermal Conductivity [W/mK]	Temp. [K]	Specific Heat [J/kgK]	Temp. [K]	Thermal expansion coefficient [m/mK]
293	11.4	293	427.14	366	1.28E-05
373	12.5	373	441.74	477	1.35E-05
573	14	573	481.74	589	1.39E-05
773	15.5	773	521.74	700	1.42E-05
973	21.5	973	561.74	811	1.44E-05
1000	21	1173	601.74	922	1.51E-05
1200	25	1623	691.74	1033	1.60E-05
1500	30	–	–	–	–

Table 5
Temperature based mechanical properties [54].

Temperature [K]	Young's Modulus [GPa]	Poisson's ratio	Yield strength [MPa]
294	208	0.3	1172
366	205	0.3	1172
477	202	0.3	–
589	194	0.3	–
700	186	0.3	1089
811	179	0.3	1068
922	172	0.3	1034
1033	162	0.3	827
1144	127	0.3	286
1227	17.8	0.3	138

the parts with respect to the nominal value which is taken from the surfaces of the CAD file as a reference. A color scale bar of ± 0.1 mm was chosen to give a good indication of which surfaces that show the largest deviation from the nominal CAD geometry. For all samples, the surface connected to the baseplate shows a large displacement. This is due to the fact that in the printing process, 0.4 mm thick layers of material are added to all the surfaces connecting to the baseplate to compensate for the material loss in the W-EDM process. This will induce a slightly high displacement due to the limited control of removal from the baseplate, e.g. more or less material loss than intended.

All the results from 3D-scanning of the parts after removal from the base plate could give indications of how the residual stresses in the part will look. If a part shows high displacement after the cut-off, this could indicate that the part has relaxed and the residual stresses in the part are lower.

For the 45B sample, the red and blue color indication can be seen on the surface along the edge of the sample, see Fig. 10. On the lower image (Fig. 10b), the surfaces facing left have a positive displacement, and the surfaces facing right a negative displacement. The larger outer surface of the sample shows less displacement. This could indicate that the part is slightly distorted in one direction.

For the VB sample, no significant displacement can be detected, see Fig. 11.

For the HB sample (see Fig. 12), one can see a clear displacement of the part bending upwards in the build direction which can be due to the large surface area connected to the base plate during the print process. After the cut off from the base plate, it will lead to the relaxation of the printed part and therefore displacements.

The measured displacement from the laser scan can be influenced by surface roughness, incorrect adjustment of the laser for the melt pool size of the outer contour causing parts to be bigger/smaller than the nominal one. So, the distortion results from the 3D-scanning can only be used as indicative measurements to be able to support the outcome from the residual stress measurements.

5.2. Residual stress distributions

As mentioned earlier, the diffraction measurement relies on the reference stress-free lattice spacing to calculate the strains. Measurements on the 3 top rows (see Fig. 6) did not reveal obvious position dependency and thus the average value of all these measurements, $d = 1.082836 \text{ \AA}$, was used as the starting point for the stress calculation. However, it was observed that using the average value of d_0 from measurement did not provide sufficient accuracy with regard to stress (force) balance in the cross-sections for all the samples (e.g. balance of σ_y for cross-section C3, σ_z for cross-section C1, etc.). This can be due to the sensitive nature of d_0 with regard to microstructure variation with different build direction. Applying stress (force) balance on all cross-sections for all the samples, a new stress-free reference of $d_0 = 1.082615 \text{ \AA}$ was found, with which the average net stress of the cross-sections is not more than ± 50 MPa, which is consistent with the accuracy of our measurements.

The results from the neutron diffraction experiment are plotted as contour maps of stress components (σ_x , σ_y , and σ_z) in Figs. 13–15 for each cross-section and each build orientation measured. Uncertainties in the measured stress values were evaluated using standard deviations obtained from diffraction peak profile fitting. They are typically in the range of ± 50 MPa. From the plots, it can be seen that RS, in general, changes from compressive inside the part to tensile near the surface of the part. Further, with different build orientations the level of RS changes. The magnitude of RS in both compression and tension was the largest for the part which was built vertically (VB) (see Fig. 14) and lowest for the part which was built horizontally (HB) (see Fig. 15).

5.2.1. Stress distribution for 45B

For the 45B part (Fig. 13), in general, the stresses σ_y and σ_z have a larger compressive zone in terms of area and it shows highly concentrated compressive stress at the center of the cross-section.

As for cross-section C1, σ_z the normal stress component to it has concentrated compressive stress at the center of the cross-section and it has a sharp gradient when it changes to tensile stress from compressive towards the surface at $Y = 0$ (denoted as Y0). This is very similar to the distribution of the corresponding normal stress component σ_y for the C4 cross-section which shows high-stress values. The difference in magnitude can be attributed to the change of part geometry with fillet present at $Y = +20$ (denoted as Y20) of C1. Further, the value of stress σ_z for C1 measured near the surface Y0 is double the value measured near the surface at Y20. This asymmetry in the stress distribution for σ_z can be partly due to the fillet present at Y20 surface and difference in heat dissipation from Y0 and Y20 surface, with the former being attached to the support structure and the latter being in contact with loose powder. For the stress component σ_y , compressive stresses are present at the center with a value of approximately -50 MPa without any significant difference in magnitude of the stresses measured near the surface Y0 and the surface Y20. Also, σ_x shows no such concentrated zone of compressive stress at the center of the cross-section. This through-

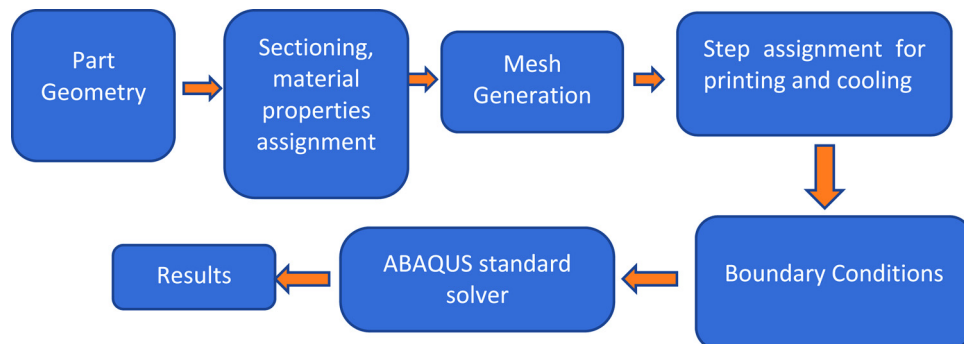


Fig. 9. FEM model workflow.

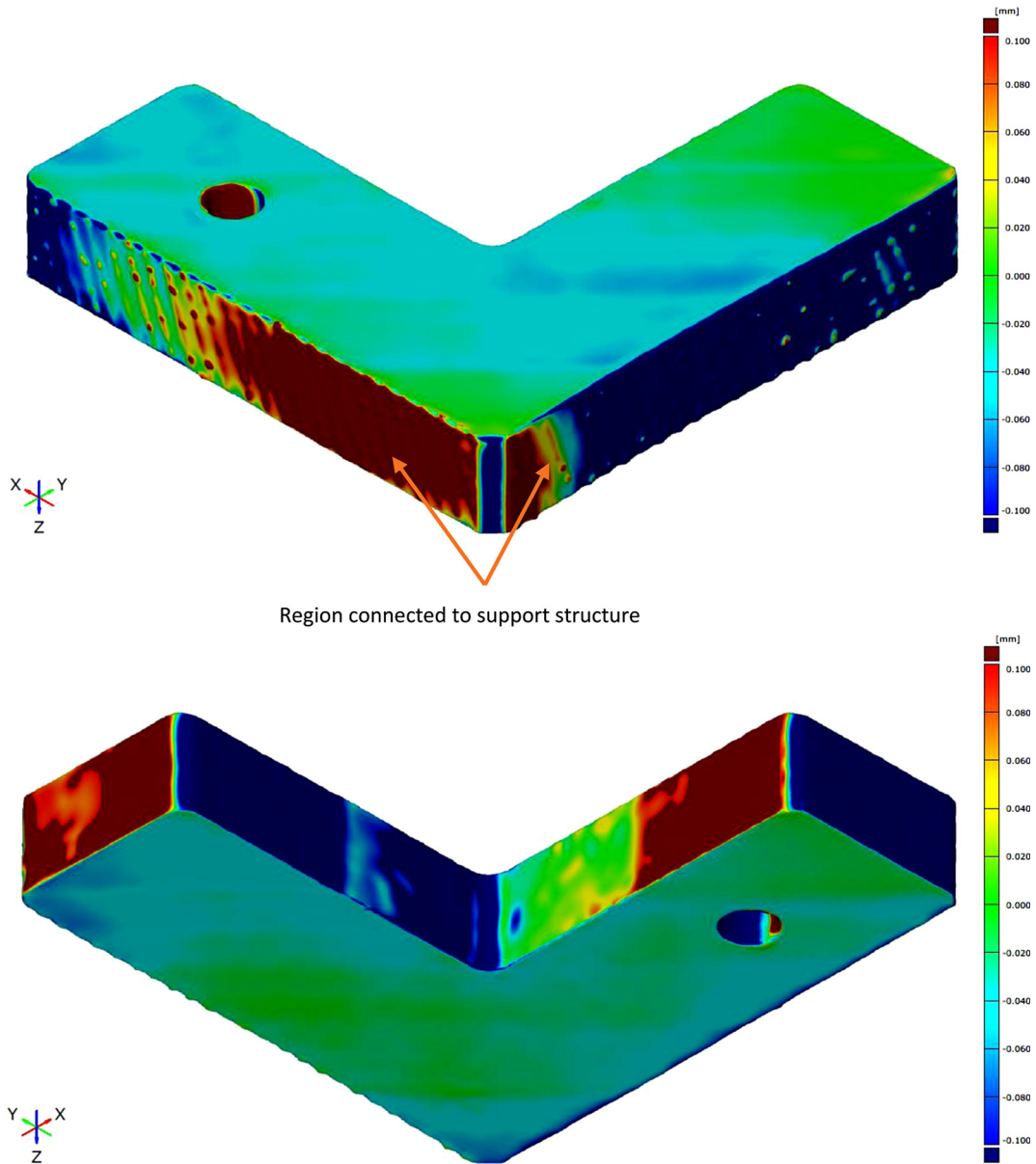


Fig. 10. Displacement of 45B sample after removal from base plate.

thickness stress component has the lowest values for both tensile and compressive stress in comparison to the other two components of stress.

When comparing cross-sections C1 and C2, it can be seen that the presence of a hole in the C2 cross-section caused larger compressive stress near the hole compared to the C1 cross-section in all three directions. Especially for σ_z in the C2 cross-section, significantly higher compressive stress near the hole in the upper half of the cross-section (around -800 MPa) when compared with σ_z of C1 cross-section and σ_y of C4 cross-section was observed. From the plots, we can also see that the stress distribution is not symmetric for C2 on both sides near the hole. This can be attributed to a combined effect of build direction and presence of support structure on one side of the cross-section (at Y0). As the cross-section measured is tilted 45° to the build direction, thermal and mechanical boundary conditions vary along the Y and Z axis, leading to the asymmetric stress distribution in these axes. It can also be seen that the presence of hole has enhanced this asymmetry.

For the C3 cross-section, we also see a larger compressive stress area for the σ_y and σ_z components. Moreover, the stress component σ_y changes to tensile stress with a sharp gradient towards surface Z = 0 (denoted as Z0). The C3 cross-section has a very elongated section of

compressive stress σ_z at the center of it, whereas a small concentrated compressive stress of σ_y is seen between 7–10 mm away from the surface Z = +20 (denoted as Z 20). Likewise, for σ_y , we can observe compressive stress above the hole as seen for σ_z for the C2 cross-section.

For the cross-section C4, the normal stress component σ_y has concentrated compressive stress at the central portion which was not observed for the other components σ_x and σ_z . Also, it has a similar distribution tendency as the normal stress component σ_z in the C1 and C2 cross-sections but with a smaller difference between stresses near the surface Z0 and surface Z20 than that for the cross-section C1 and C2 (Y0 and Y 20). Comparing with C2 which is at the same distance from the origin, we can see that the presence of the hole affects mostly the stress component normal to the cross-section.

Further, the stress component in the thickness direction (σ_x) is in general, lower than the other two components, σ_y , and σ_z for all the cross-sections. The side of the part connected to the support structure has higher values of stresses compared to the upper half of the cross-section and the gradient of stress change is also sharper than on the other side. Obviously, such an asymmetric stress distribution is related to the different boundary conditions for cooling and different

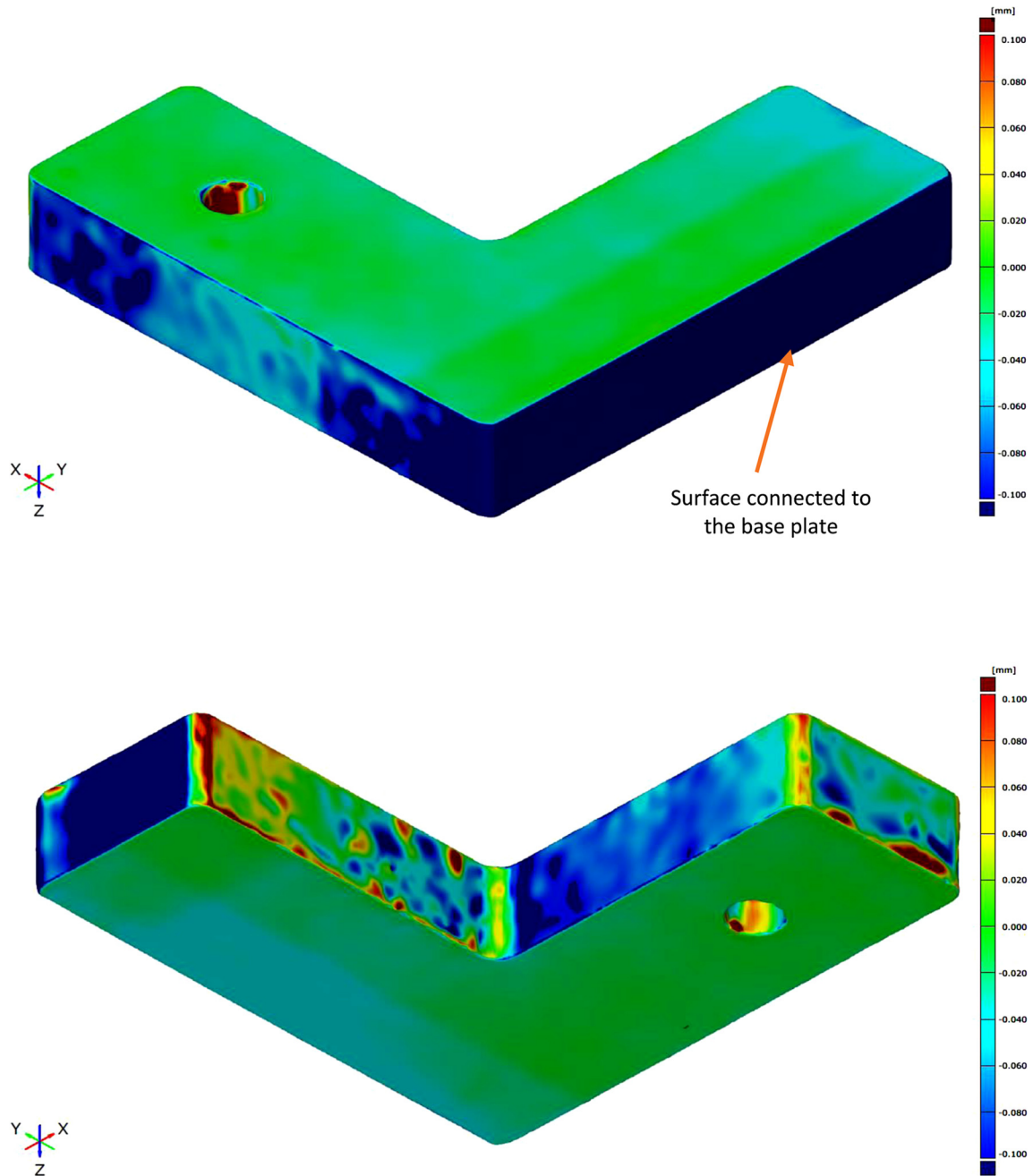


Fig. 11. Displacement of the VB sample after removal from base plate .

mechanical constraints during cooling as explained earlier. The C2 and C4 cross-sections are at the same distance from the origin, and comparing the stress distribution, we can see that the normal stress component σ_y of C4 is more symmetrical than the normal stress component σ_z of C2. Such a difference is originated from the presence of the hole at C2.

5.2.2. Stress distribution for VB

For the part built in the vertical orientation (VB), higher stresses appear in the Z-direction (building direction) for all cross-sections when compared to the other two directions, and the tendency of the compressive to tensile stress from the center to surface has also been observed as shown in Fig. 14. For the cross-section C1, σ_z shows a big compressive area in the bulk of the part and a higher level of tensile residual stress near the surface Y0 in comparison to the stress levels near the surface Y20. This sort of asymmetry in stress distribution near

the surface Y0 and surface Y20 can be linked to the fillet that is present on the Y20 surface. Both σ_y and σ_x for the C1 cross-section show a lower level of the stresses in both compression and tension. For the cross-section C2, due to the presence of the hole, high compressive stress is observed near the hole. The compressive stress zone is similar on both sides and here also σ_x and σ_y are lower than σ_z . There is a slight asymmetry for σ_z for C2 on the opposite sides of the hole especially near the surfaces Y0 and Y20. This can be attributed to the combination of the presence of the hole which is printed with the support structure and measurement uncertainties. For the C3 cross-section, an elongated zone with high compressive stress is observed in the lower half of the cross-section for σ_z . For σ_y also a compressive stress zone is observed but with a lower magnitude of the stresses. The stress component σ_x is lower than the other components and does not show a compressive zone in the bulk. For the cross-section C4, the stress levels for σ_z is generally lower than for σ_z in the C1, C2, and C3 cross-sections. The stress components

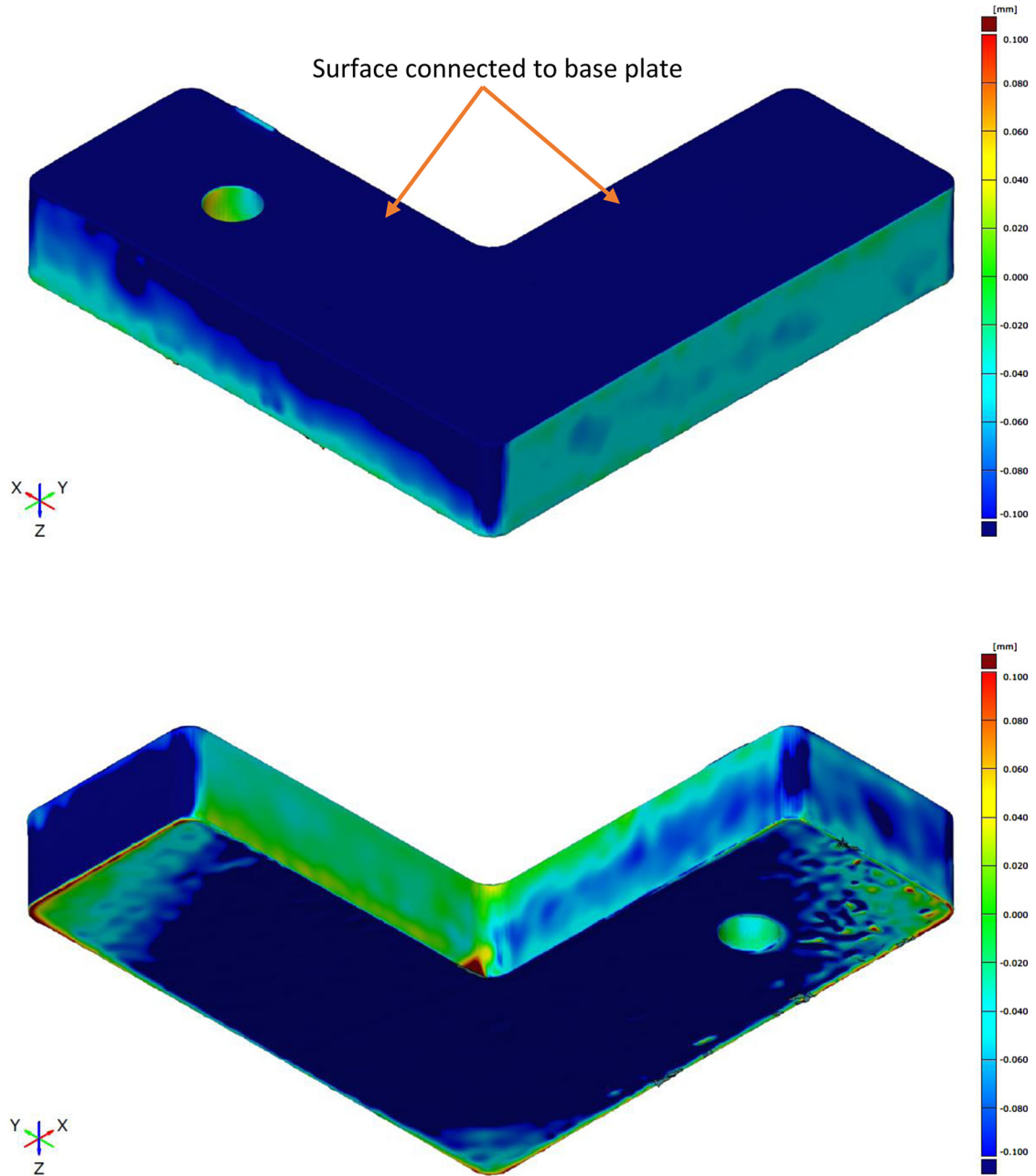


Fig. 12. Displacement of HB sample after removal from base plate.

σ_x and σ_y also have similar distribution as of the C1 cross-section.

For this build orientation, the surface at Z0 for the C3 and C4 cross-sections were connected to the base plate and here we can observe slightly higher σ_y but lower σ_z in comparison with other cross-sections. As the C1 and C2 cross-sections are far away from the base plate region, the influence of the removal will not be significant due to which the levels of the σ_z near the Y0 and Y20 surfaces are of the order 4 times higher than those measured for the C3 and C4 cross-sections near Z0, Z55, and Z0, Z20 respectively. This sort of stress relaxation can be observed for the normal stress components σ_y of C3 and C4 when compared to the normal stress components σ_z of C1 and C2.

5.2.3. Stress distribution for HB

For the part built-in the horizontal orientation (HB), the stresses in all 3 directions are lower than those of the other two build orientations and keep a similar trend of the stress distribution as others and are presented in Fig. 15. Here the stress components in all the 3 directions

show a gradual change from compressive to tensile rather than a steep gradient in a small interval, as seen in the other two build orientation. For the C1 cross-section, the σ_y component is compressive in the bulk with levels of around -200 MPa and tensile near the surface at $X = +5$ with stress levels near 300 MPa. The σ_x and σ_z components have a lower magnitude than σ_y . For the C2 cross-section, due to the presence of the hole, there are slight compressive stresses formed near the hole, especially for σ_y . For the C3 cross-section, σ_y shows an elongated compressive zone as seen for the other build orientations before but with lower levels of the stresses. The highest levels of the tensile stresses were measured for C4 in the Z-direction. For the cross-sections, C2 and C4 which are at equal distance from the origin, the values of the normal stress component σ_y for C4 and σ_z for C2 near the surface at $X = -5$ and $X = +5$ are in the range of 300 MPa which also confirms the symmetry expected in these cross-sections.

Here the surface at $X = -5$ was connected to the base plate, and from the measurement, it can be assumed that removal from the base

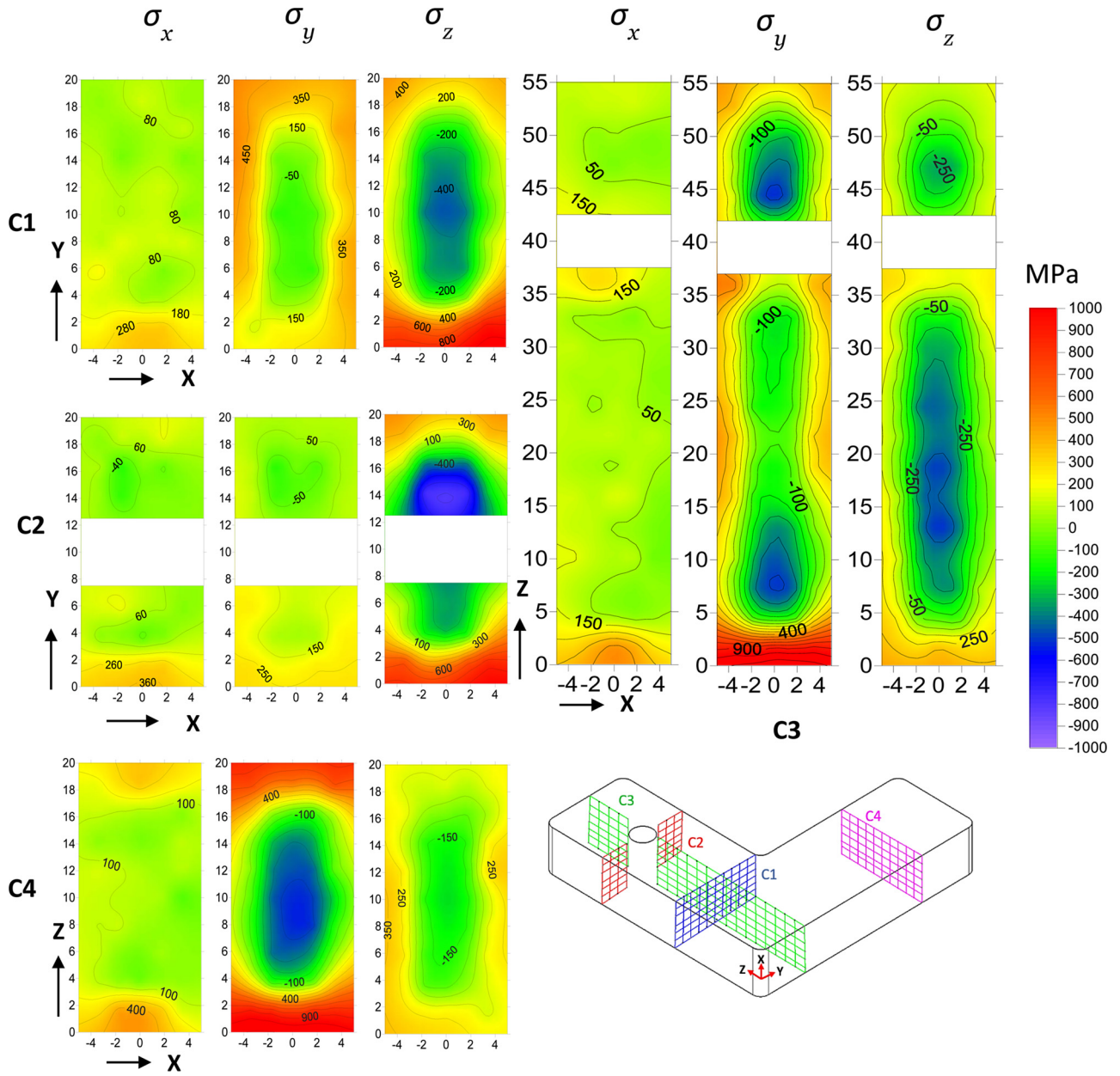


Fig. 13. Stress distribution at different cross-section for 45B sample.

plate led to relaxation for the whole cross-section and the deformation of the HB sample (See Fig. 12) is indicative of that. As the thickness of the sample was 10 mm, the influence from the base plate removal was significant for all the surfaces, which is evident also from the comparison of stress distribution in respected cross-sections with the other two samples, 45B and VB.

5.2.4. Surface stress measurement

Surface stress measurement was performed on the side face at cross-section C4 for the 45B samples at the center of the cross-section (L1) ($X = 5$, $Z = 10$) and at the point 3 mm away ($X = 5$, $Z = 3$) from the shorter edge (L2) (see Fig. 8). From surface stress measurements, we observed that the stresses in the Z direction are in tension with the magnitude of 218 ± 48 MPa at L1 and 414 ± 61 MPa at L2. For the Y direction stress of -81 ± 47 MPa was observed at the L1 and a stress of 383 ± 55 MPa was observed at L2. These measurements are consistent with the data from the neutron diffraction experiment. Combined neutron and X-ray data at both locations can be seen in Figs. 16 and 17.

The points at the surface are from X-ray and the rest of the points inside the material are from neutron diffraction.

5.3. Results from the FE simulation

Simulations were performed to predict the residual stress in the 45B sample. As the sample built in 45-degree orientation (45B) was built with support structures of low density, the properties of the support structure were scaled down to 50 % for the density and 25 % for the mechanical properties compared to the original values.

From the simulation results, residual stress data were extracted for the cross-sections where the neutron experiment was performed. Data were extracted at the integration points of the elements. The predicted 2D residual stress distributions as seen in Fig. 18 are similar to those obtained experimentally. Asymmetric stress distribution of σ_z at the C2 cross-section was also revealed. On the other hand, the predicted stresses near the surfaces that were attached to the base plate are slightly lower than experimentally measured values. Non-conformity of

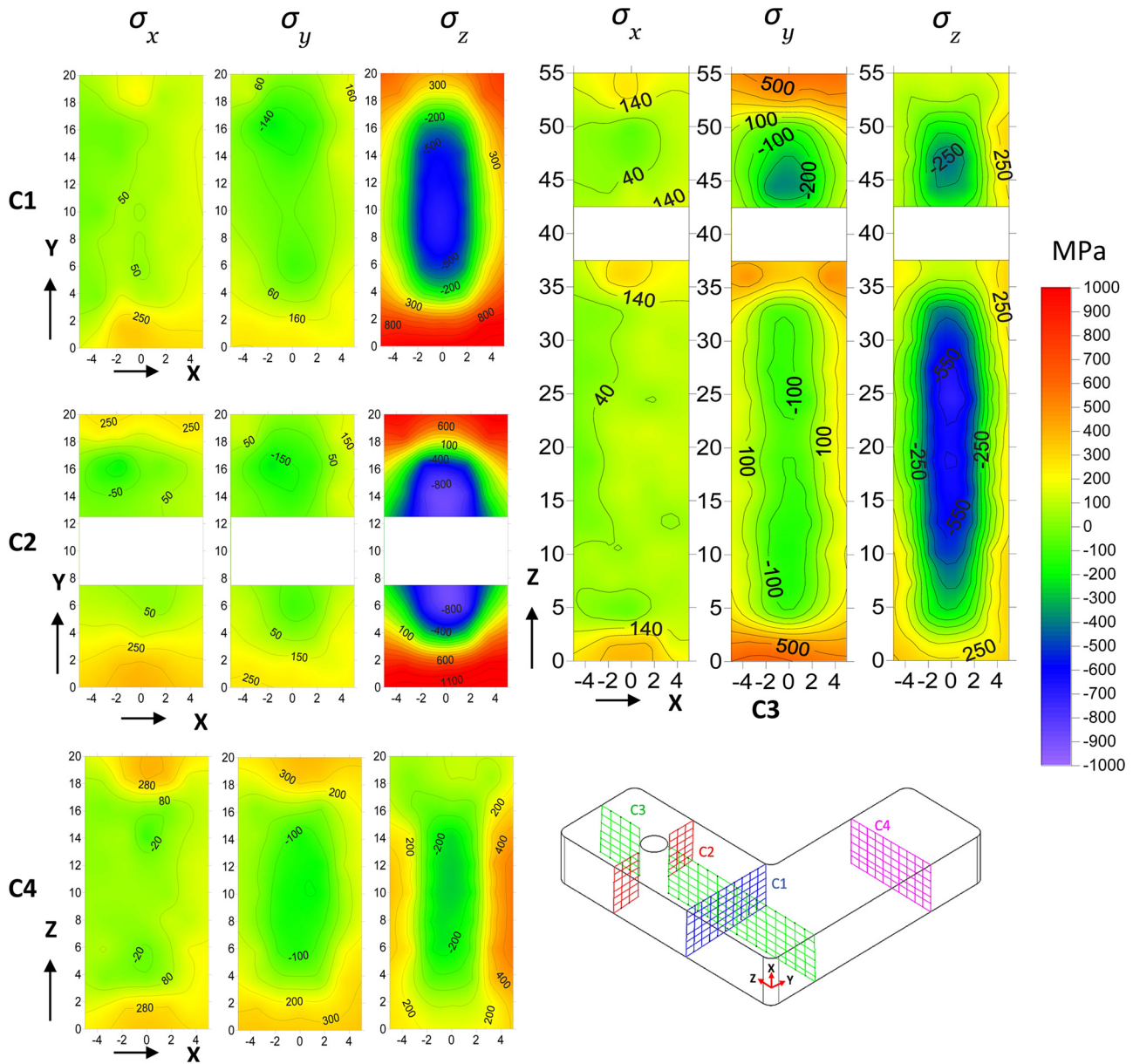


Fig. 14. Stress distribution at different cross-section for VB sample.

the stress levels from the measured ones can be for instance attributed to the assumptions for the material behavior and uncertainty in material properties used for the simulations. For example, the material properties used for the simulation are for aged wrought In718 but the as-built component will not have the same properties as aged material. Also, it was idealized that the material is isotropic but in reality, it has been observed that AM material shows anisotropic behavior [55]. Other idealized and simplified assumptions for heat dissipation conditions may also play a role in the deviations between prediction and measurement. Nonetheless, the goal of the modeling is to identify critical areas with regard to RS, and the model indeed predicts the trend of the stress reasonably well in comparison with the experiments.

6. Discussion

The residual stresses in the as-built state (in removed from the base plate condition) were measured by neutron diffraction for L-shaped parts printed with three different build orientations. The shape, size, and features of the part were to emulate the real-world scenario, where

RS are vital for the part performance. It was observed that in all build orientations, the stress trend was compression in the bulk and tension near the surface as reported in previous works [40,56–59]. A high level of tensile residual stress on the surface can lead to cracks during printing while specific stress distributions can lead to deformation or distortion and these two effects of RS can make the component unusable. Cutting a part off the build plate is expected to cause the relaxation of residual stresses that developed during the build process. Results from part distortion measured by the 3D surface scan technique show that, global residual stress distributions before removing the part can be inferred qualitatively. On the other hand, with FEM simulations closer evaluation of the non-relaxed residual stresses can be made. The results of the simplified FEM simulations show a similar trend and in a good qualitative agreement with the experimental data, which in this way allowing to test and verify the simplified model for stress prediction. The magnitude of the stresses predicted are higher than the measured ones due to the material properties used and the simplification of the boundary conditions as well. Despite the observed similarities in the type of stress distribution (tensile stress on surface vs

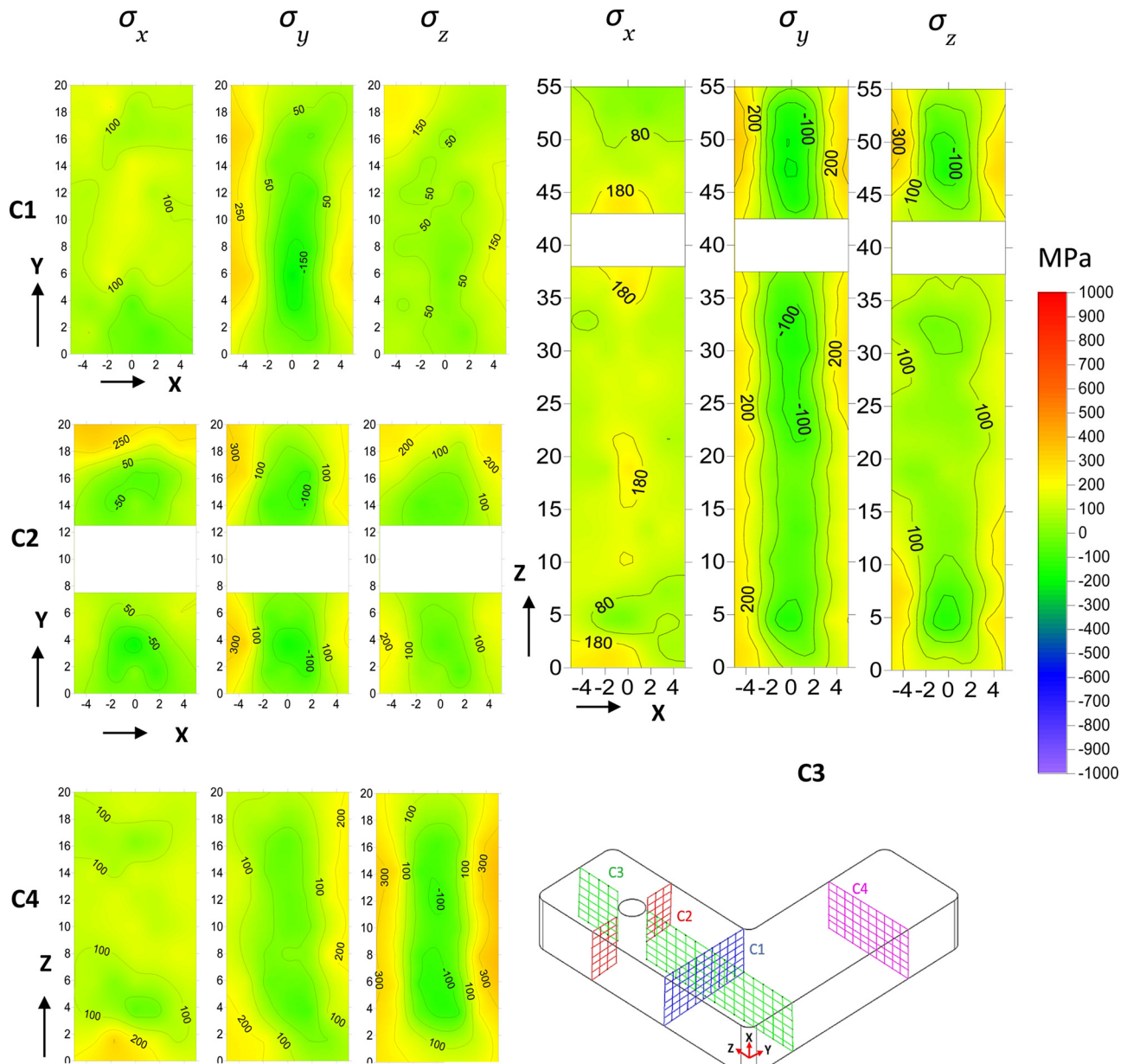


Fig. 15. Stress distribution at different cross-sections for HB sample.

compressive stress in core), the magnitude of the RS distribution for all the cross-section measured is different for different build-up directions. The lowest magnitudes of RS were achieved for the HB part.

The overall contribution to the residual stress distribution in the as-build samples comes mainly from two constituents that can be distinguished. The first (RS1) is the RS developed due to interaction of the build material and the base plate, while the second (RS2) is the RS that are self-equilibrated within the build component when it is removed from the base plate. During AM process, the combination of the both is of interest, especially when mechanical and/or thermal properties of the substrate and deposits are different because the superposition of both kinds of RS1 and RS2 can lead to delamination of the part from the base-plate, a distortion and a loss of dimensional tolerance. The RS2 is important for the in-service life and performance of the AM part, e.g. affecting fatigue performance. They can be largely considered separately as follows.

6.1. RS1

The influence of build direction on RS1 can be derived by comparing the measured deformation due to the removal of the parts from the base plate (see Figs. 10 to 12). The deformation measurement indicates that the 45B sample has the lowest amount of overall deformation compared to other build orientations. As the contact area between the part and the base plate is small and the support structure was much more compliant, the constraint of the base plate to the thermally induced deformation of the part during the building is low. Analogously, the HB part, which has the largest contact area with the base plate indicated the highest degree of deformation at the upper and lower surface due to a strong interaction with the base plate (see Fig. 12). In between, we have the VB sample, which in reality has the least contact area with the base plate and in theory, must have high deformation at the lower areas near the base plate (see Fig. 11). The values of the stresses near the areas that were connected to the base plate were lower than for the areas that were connected to the support structure in the 45B sample as indicated by the displacement map in

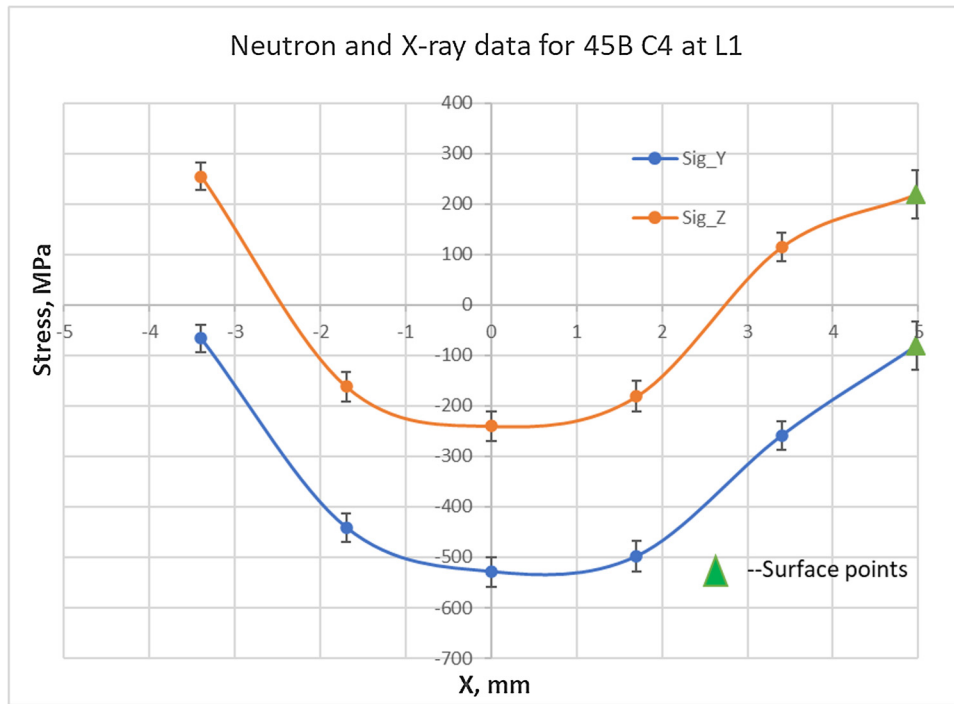


Fig. 16. Combined neutron and X-ray data for 45B C4 at L1 (Note: surface points are from X-ray measurement).

Fig. 10. While printing complex parts this has to be taken into account because the stress relaxation during the removal can lead to significant distortion or even cracking of the part which will make it unusable. Secondly, this effect will be particularly significant for a thin-walled structure when printing in horizontal orientation can lead to significant bending of the part. Even though the RS in the as-built state for the parts with orientation VB and 45B is higher than in the HB part, but due to lower distortion, it can be a better choice for the print. The HB orientation can also be exploited if we use some additional dummy layers between the base plate and part with reduced mechanical properties.

6.2. RS2

This part of the RS is developed during the printing processes when for each newly depositing layer the previously deposited layers play the role of constraint. They are balanced within the samples and are retained in the sample after the cutting. The main principles and general mechanism of this stress formation are well known [16]. During the printing process of selective laser melting a steep gradient in temperature is formed due to the rapid cooling, this will cause rapid shrinkage leading to tensile stresses in the new layers [18] and thus the RS variation in the build direction. This can be seen for the VB sample

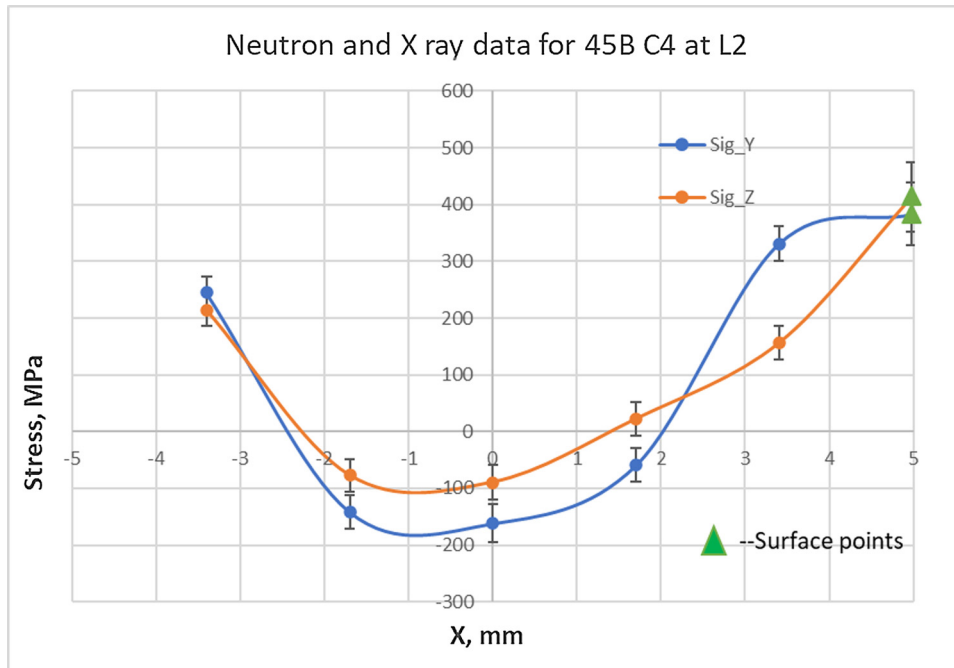


Fig. 17. Combined neutron and X-ray data for 45B C4 at L2 (Note: surface points are from X-ray measurement).

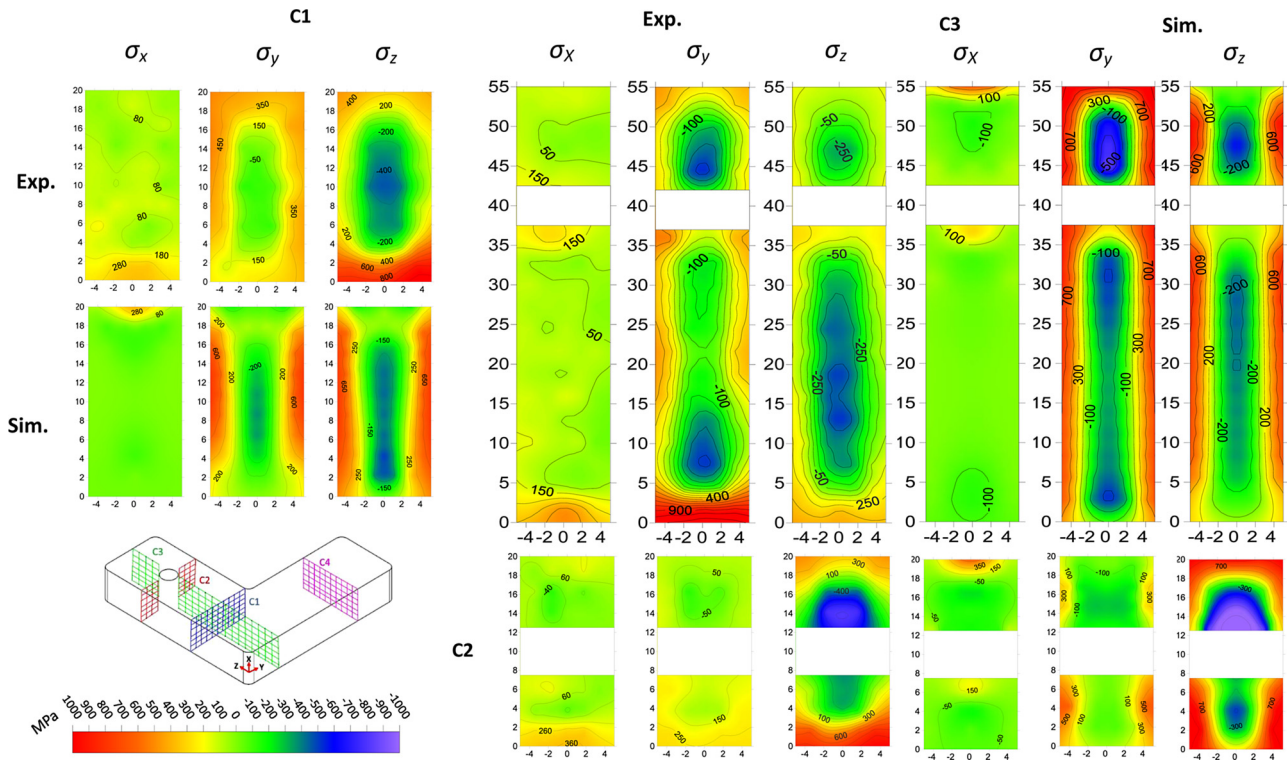


Fig. 18. Comparison of measured stress distribution with FEM results for 45B sample at different cross-sections.

as σ_z and for HB sample as σ_x which are presented in Figs. 14 and 15 for different cross-sections respectively. Furthermore, the area near to the surfaces will have a higher cooling rate due to the additional boundary condition of conduction to the loose powder bed rather than the bulk which has conduction through the base plate as the main mechanism of the heat transfer, which results in RS distribution transverse to the build direction. These RS distributions can be seen for the 45B, VB, HB samples in Figs. 13–15 respectively. One way to reduce the temperature gradient between these two subsequent layers will be to rework the contour at the end with the laser and alternating the contouring and hatching between the layers might also help to lower the effect of the temperature gradient. Detail investigation regarding the influence of contouring parameters on the RS seems necessary in order to fully understand the effects near the surface.

The simplified approach used in this study for the simulation of RS has been able to predict the stress distribution tendency reasonably well with respect to the experimentally obtained stress distributions. The simplified model can be incorporated into the design process before printing the part as the model presented here can predict critical areas in the complex structure and contribute to less scraped material. Further, this model can be coupled with topology optimization to get the best geometry based on the loading conditions as well as RS. This simplified method can be further developed to incorporate more complex features like a moving heat source, mesh regeneration, complex material models, etc. to name a few depending upon need. Incorporation of these features might give more accurate predictions of the stress levels but will also increase the computational time significantly.

7. Summary and conclusion

This work presents non-destructive investigations based on the neutron diffraction technique to quantify the residual stresses in additively manufactured L-shaped Inconel 718 parts, printed in three different build orientations by laser-powder bed fusion. Part deformation due to removal from the build plate was also measured by the GOM

Atos Triple Scan III system. Extensive strain mapping in 4 selected cross-sections in the specimens removed from the base plate allowed us to gain further understanding of the influence of the build orientation, support structure, and stress concentrations on the final outcome of residual stress. Results from the experiment were compared to the results from a simplified thermomechanical FE simulation. Based on the results, the following conclusion can be drawn:

- The Neutron diffraction technique can be used to measure the residual stresses present in additively manufactured parts accurately and with sufficient spatial resolution.
- The measurements typically show a residual stress distribution changing from tensile near the surface to compressive in the core of the specimen.
- The magnitude of the residual stresses is in the order of horizontal built < 45° built < vertical built. In all the samples Z-component of stress is strongest and this also corresponds to build direction for VB sample.
- The horizontal sample has the highest degree of deformation during the removal of the parts from the base plate, while Vertical samples and part built at an angle of 45° with support structures have less deformation while removing them from the base plate.
- Build orientation has a significant impact on the residual stress magnitude. A careful selection of the printing orientation is important. For example, horizontal built orientation can be useful to print parts with significant thickness, as for the thin-walled samples printing with support structures at an angle or print vertically with support structure will be beneficial.
- The simplified FE approach for the prediction of the residual stresses is in good qualitative agreement with the experiment and the major stress distribution features such as the hole, surface connected to base plate and trends are reproduced. Further improvement of the model is necessary to close the gap between the measured and predicted values of RS.
- Further investigations regarding the influence of different scanning strategies and laser power on residual stresses will be necessary to

get a better understanding of the complex nature of the relationship between different processing parameters and residual stresses.

Authors statement

Prabhat Pant was involved in conceptualization, methodology, investigation, formal analysis and use of software tools, data presentation, writing- original draft, reviewing and editing.

Sebastian Proper was involved in sample preparation (resources), planning and investigation of the experimental result together with input in writing of the draft.

Valdimir Luzin was involved in planning the neutron experiment, investigation, formal analysis and writing initial draft and revised version together with providing resources.

Sören Sjöström, Kjell Simonsson, Johan Moverare were involved in methodology, conceptualization, providing resources in terms of software analysis, validation of the FE model and, data presentation, supervision as well as reviewing the original and revised draft.

Seyed Hosseini was involved in providing samples and supervision and feedbacks on the work and reviewing the original and revised draft.

Victor Pacheco was involved in performing neutron experiment and initial analysis of the neutron data.

Ru Lin Peng was involved in conceptualization, methodology, supervision, funding acquisition, investigation, providing resources, project administration, data analysis, data presentation, and writing original draft together with reviewing the revised version.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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